LAB #12

Adv. CiC
Spring 2026

Agenda

1. Infra / Compute 101
2. Networking 101
3. Cloud 101
4. Lab 12 Exercise

Housekeeping

Due this past week

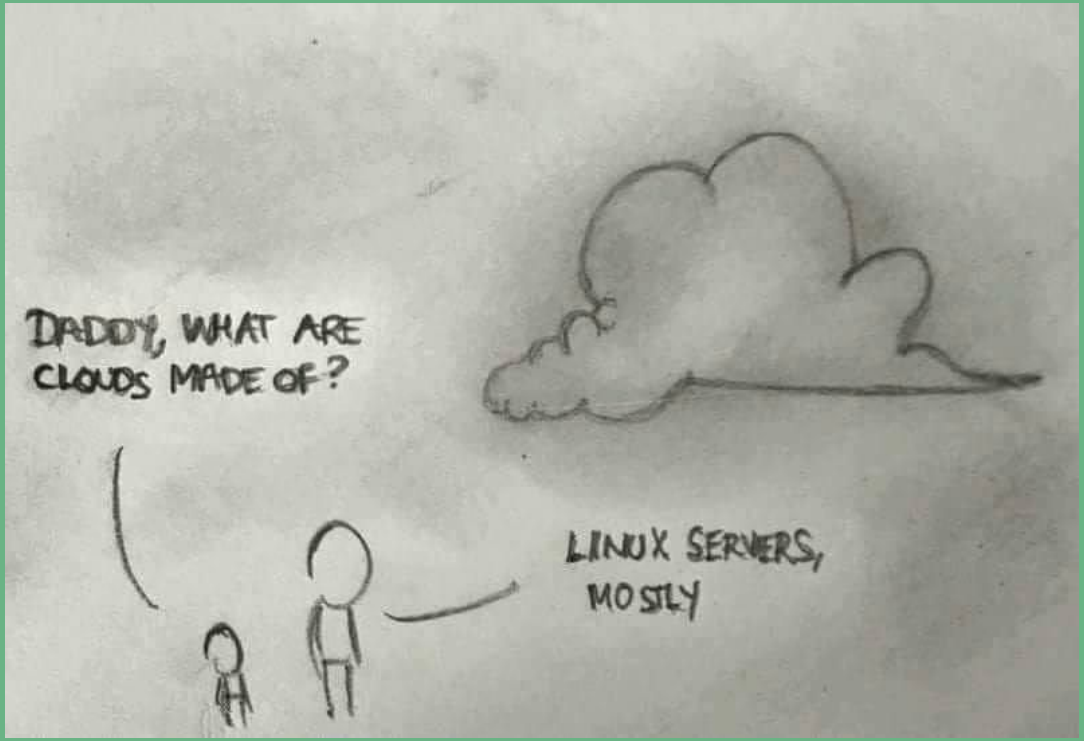
- Readings
- Lab 11

Upcoming

- Readings - **4/16 @ 3pm**
- **Part 6 - 4/16 @ 3pm**

Announcements

1. Grading catch-up (still ongoing...)



Questions
(on Lectures, Labs, or Readings)?

Infra / Compute 101

Basic Terminology

- **Server:** A computer that provides resources or services to other computers over a network
- **Workload:** The application or task you're running on infrastructure (a website, ML job, database, etc.)
- **Compute:** Processing power (CPU cycles) used to actually execute your workload
- **GPU:** A processor with thousands of small cores for parallel computation - needed for AI/ML

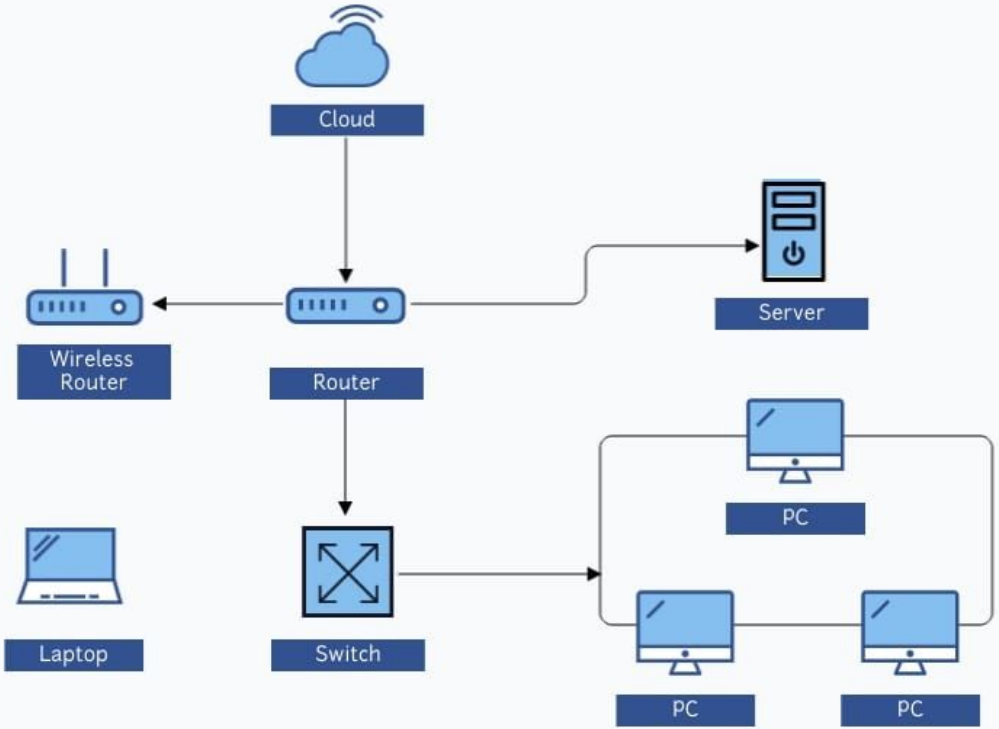
Basic Terminology (cont.)

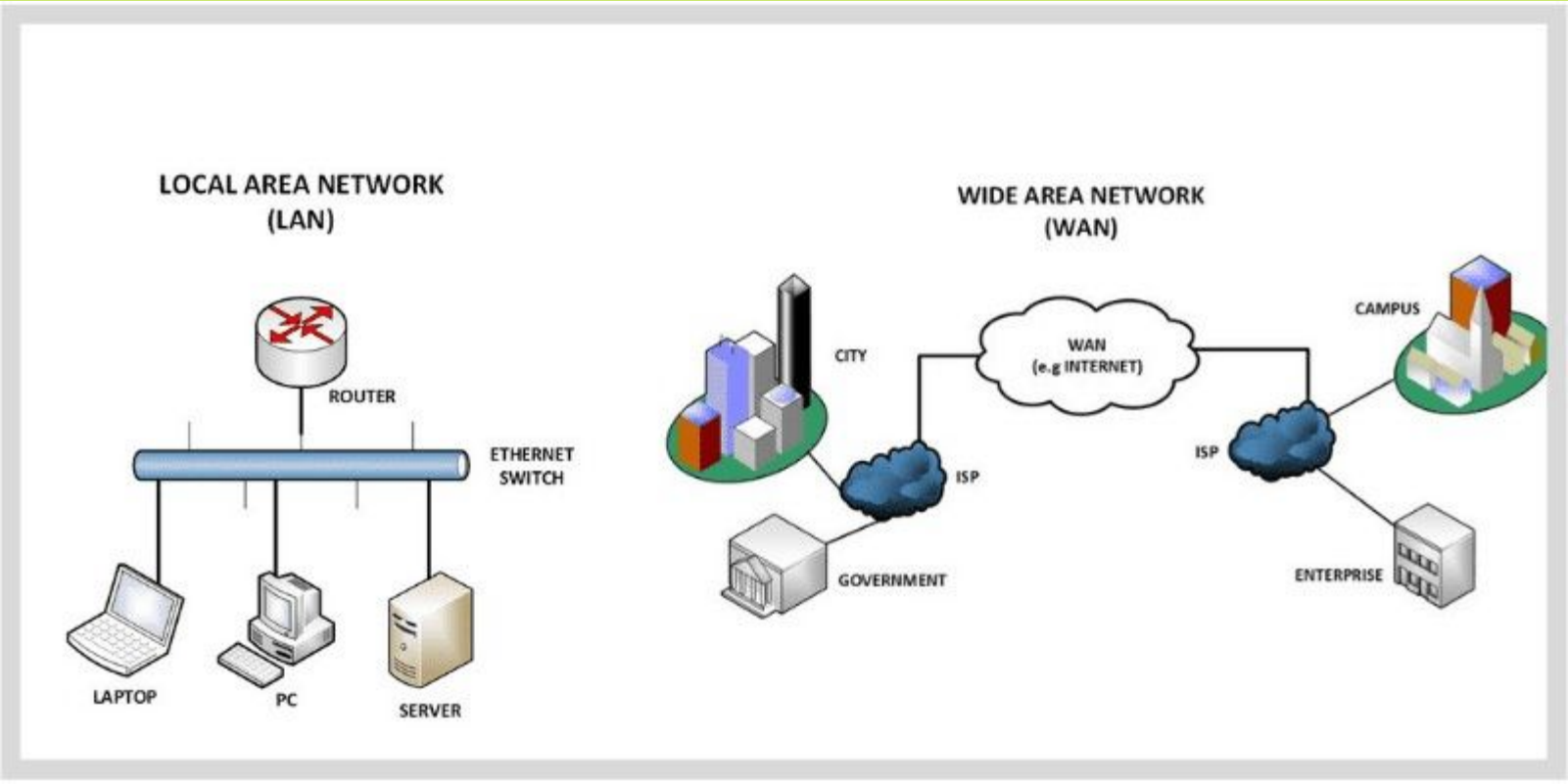
- **Memory:** A broad term for any system that stores data — exists on a spectrum from fast/temporary to slow/persistent
  - **RAM (Random Access Memory)**: Short-term memory — fast, temporary storage holding data your program is actively using, wiped on restart
  - **Storage:** Long-term memory — persistent data that survives restarts

- **IAM (Identity and Access Management):** The system controlling who can do what to which resources

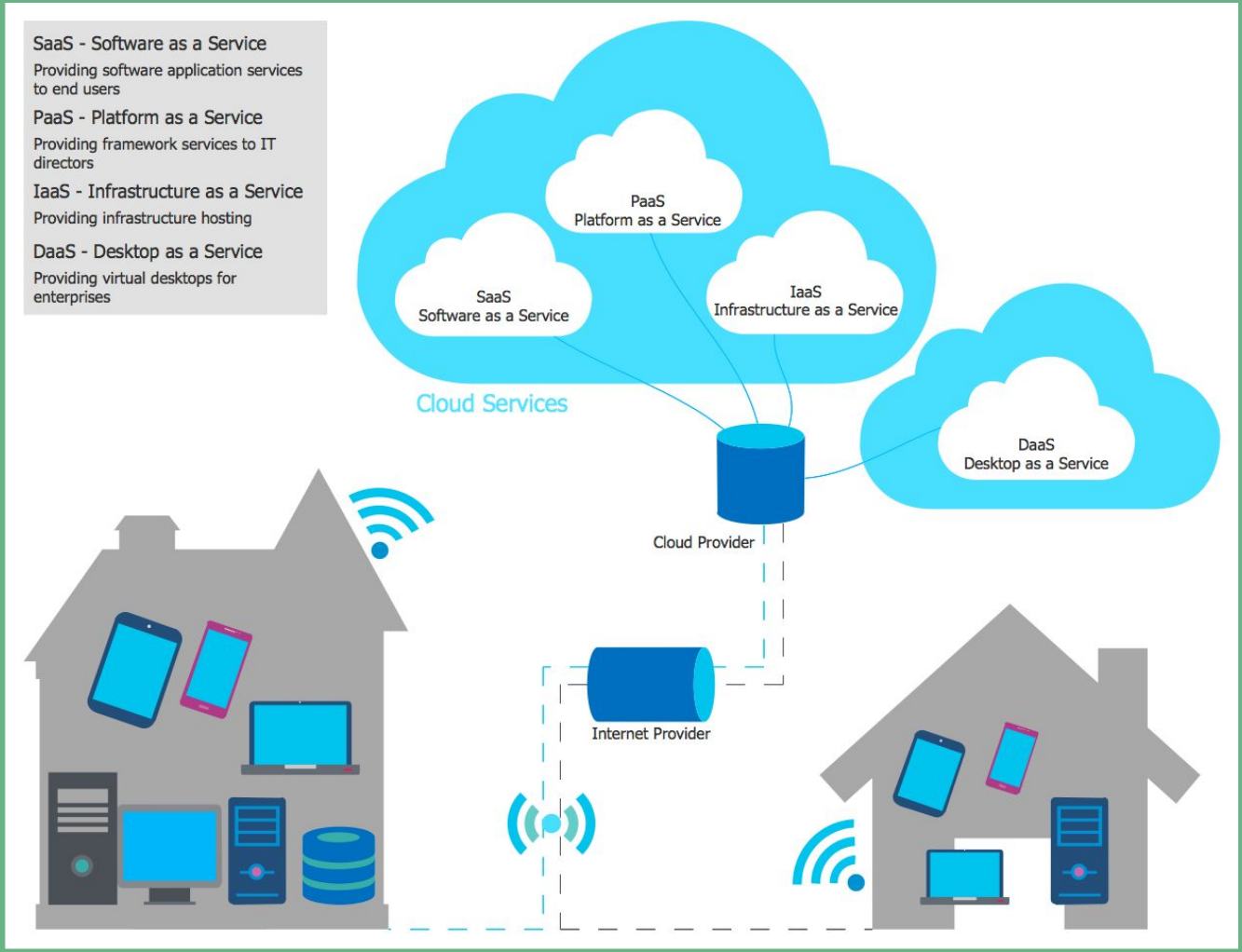
Networking 101

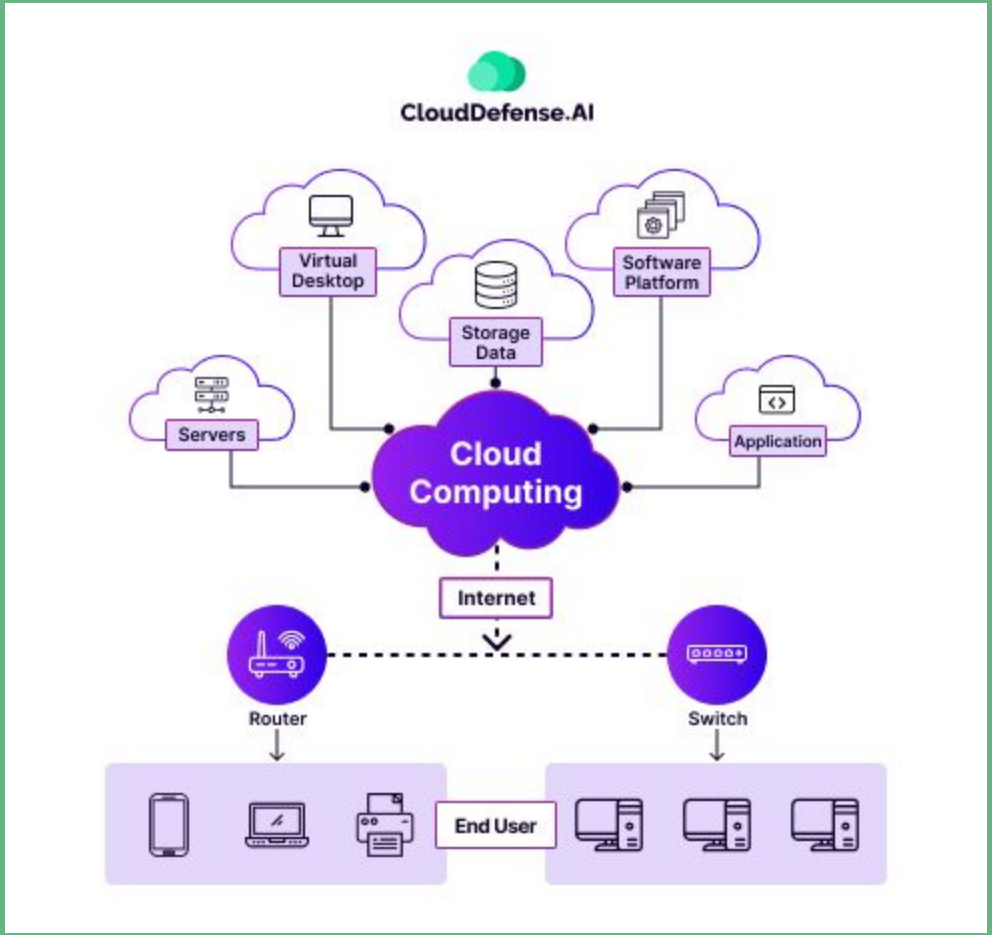
Computer Network Diagram




LOCAL AREA NETWORK
(LAN)
ROUTER
ETHERNET
SWITCH
LAPTOP
PC
SERVER
WIDE AREA NETWORK
(WAN)
CITY
WAN
(e.g INTERNET)
CAMPUS
ISP
ISP
GOVERNMENT
ENTERPRISE

Cloud 101


SaaS - Software as a Service
Providing software application services to end users
PaaS - Platform as a Service
Providing framework services to IT directors
IaaS - Infrastructure as a Service
Providing infrastructure hosting
DaaS - Desktop as a Service
Providing virtual desktops for enterprises
PaaS
Platform as a Service
SaaS
Software as a Service
IaaS
Infrastructure as a Service
Cloud Services
DaaS
Desktop as a Service
Cloud Provider
Internet Provider


CloudDefense.AI
Virtual Desktop
Storage Data
Software Platform
Servers
Application
Cloud Computing
Internet
Router
Switch
End User

ATARI

EXERCISE

Lfg